Labsphere

8°/Hemispherical Reflectance Calibration Certificate

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Authorization: Sales Order:

Calibration Laboratory: Labsphere, Inc., Reflectance Calibration Laboratory
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Description of Calibrated Items
One white diffuse reflectance sample, model SRT-99-050 Reflectance Target, serial number 99AA04-1221-0838

Calibration
8°/Hemispherical (8/h) Spectral Reflectance Calibration over the range 250 – 2500nm, reported at 50nm intervals

Description of Calibration
The calibration items are measured using a Perkin Elmer Lambda 900 or Lambda 950 dual beam spectrophotometer with LambdaX 150mm InGaAs RSA integrating sphere reflectance accessories which perform the 8/h reflectance measurements. The instrument used for the measurement of the calibrated items is identified with an X in Table I below.

Table I Measurement Instrument and Accessory

Spectrophotometer	Lambda 900A S/N 101N 3060902	Lambda 950B, S/N: 950N3110101	Lambda 950C SN: 950N9031801
Accessory	LambdaX 150mm InGaAs RSA Serial No. 0508182129	LambdaX 150mm InGaAs RSA Serial No. 0508182127	LambdaX 150mm InGaAs RSA Serial No. 0508182128
Instrument Used		X	

The integrating sphere diameter is 150 mm, the sample port diameter is 25 mm, and the interior material is Spectralon. The reference standard and sample piece are placed sequentially in the sample port of the sphere. The reference beam of the instrument acts as an auxiliary beam to correct for substitution error in the sphere. The calibration of the instrument follows the NIST method of utilizing pressed polytetrafluoroethylene (PTFE) as the reference standard[1,2].

Uncertainty values take into account the uncertainties of the pressed PTFE reference standard. The 8/h spectral reflectance of the sample R_S at each wavelength λ was calculated from:

Eq. 1
$$R_S = \frac{M_S - M_H}{1 - M_H} \cdot R_R$$

where:
M_S is the instrument's relative measurement of the reflectance of the calibration item
M_H is the instrument's relative measurement of the reflectance of an open sample port and measures the effect of stray light overfilling the sample port
R_R is the 8/h spectral reflectance of the pressed PTFE reference standard.

[1] Wiedner V.R., and Hsia, J. J. "Reflection Properties of Pressed Polytetrafluoroethylene Powder", J.Opt.Soc.Am., Vol71, 1981, pp856-861

[2] Barnes, P.Y., Early, E.A., and Parr, A.C., "NIST Measuremetn Services: Spectral Reflectance," U.S. Dept. of Commerce, 1998.

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The instrument's relative measured reflectances, M_S and M_H are equivalent to:

Eq. 2

$$M_S = \frac{S_S}{S_R}$$

and

Eq. 3

$$M_H = \frac{S_H}{S_R}$$

where:
S_R is the signal with the reference standard in place
S_S is the signal with the calibration item in place
S_H is the signal with an open port

Therefore, the 8/h spectral reflectance of the calibration item R_S at each wavelength λ is equivalent to:

Eq. 4

$$R_S = \frac{S_S - S_H}{S_R - S_H} \cdot R_R$$

The final 8/h spectral reflectance is obtained by averaging the values from three scans.

Sources of uncertainty are:

- the 8/h spectral reflectance of the pressed PTFE reference standard
- nonuniformity across surfaces of the reference standard
- nonuniformity across the surface of the calibration item
- residual uncertainty in the correction for extraneous light
- nonlinearity of the instrument
- effect of wavelength error
- random noise in the measurements

The individual contributions to uncertainty are combined by adding in quadrature (root-sum-square). The quadrature sum is multiplied by a coverage factor (k) to generate the expanded uncertainty. The coverage factor is chosen to provide a confidence level of 95%. For effective degrees of freedom of thirty or greater, k = 2.0. For lower degrees of freedom a larger coverage factor is used representing the coverage factor necessary to provide a confidence level of 95% for a t-distribution with the corresponding degrees of freedom. Reference Table IV for the expended uncertainty of the calibration results.

Jess Holland

Calibrated by:
Title: Optical Calibration Technician **Jess Holland**

Approved by:

General Information Tim Smith

1. The values in Table III apply only to the central 14 mm by 6 mm area of the items for measurement geometry 8/h.

2. The report of calibration may not be reproduced except in full without the written consent of this laboratory.

3. This report must not be used to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the federal government.

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Table III. 8/h spectral reflectance R_s as a function of wavelength λ

Wavelength (nm)	Reflectance
250	0.9605
300	0.9751
350	0.9824
400	0.9874
450	0.9858
500	0.9849
550	0.9881
600	0.9880
650	0.9880
700	0.9869
750	0.9868
800	0.9855
850	0.9870
900	0.9864
950	0.9863
1000	0.9864
1050	0.9861
1100	0.9862
1150	0.9858
1200	0.9850
1250	0.9849
1300	0.9840
1350	0.9842

Wavelength (nm)	Reflectance
1400	0.9826
1450	0.9832
1500	0.9834
1550	0.9833
1600	0.9830
1650	0.9817
1700	0.9805
1750	0.9802
1800	0.9804
1850	0.9754
1900	0.9749
1950	0.9718
2000	0.9664
2050	0.9550
2100	0.9466
2150	0.9429
2200	0.9590
2250	0.960
2300	0.954
2350	0.943
2400	0.938
2450	0.935
2500	0.941

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Graph I. 8/h spectral reflectance R_s as a function of wavelength

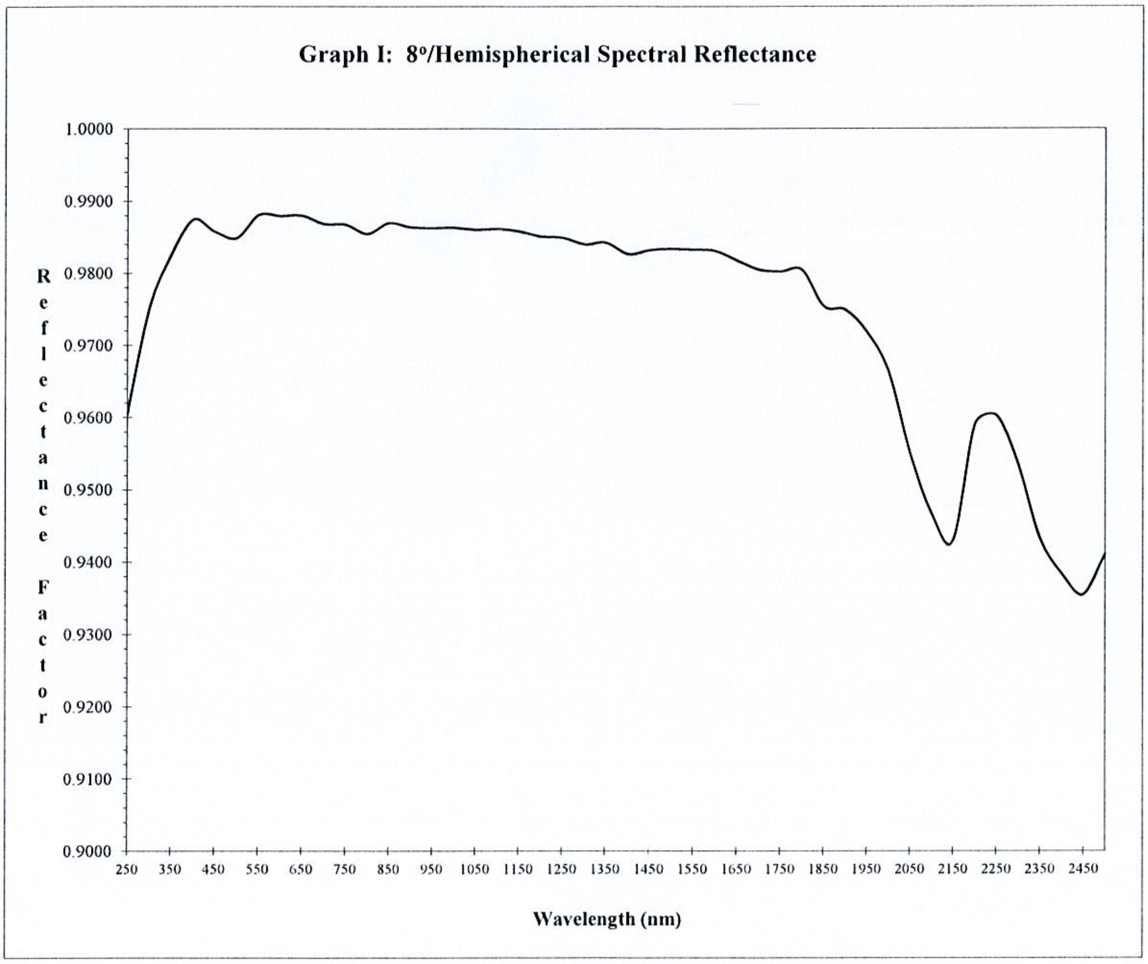

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Table IV Uncertainty contributions and expanded uncertainty (k=2) of the 8/h spectral reflectance for the measured sample

CALIBRATION AND MEASUREMENT CAPABILITIES (CMC) Notes 1,2

Measured Parameter or Device Calibrated	Range	Uncertainty (k=2) Note 3,5	Remarks
OPTICAL RADIATION			
Photometric (20/O02)			
Relative Reflectance at Wavelength Shown Below:			Relative reflectance is a dimensionless quantity
250 nm to 600 nm	up to 0.02	0.0016	
	> 0.02 to 0.05	0.0029	
	> 0.05 to 0.10	0.012	
	> 0.10 to 0.20	0.012	
	> 0.20 to 0.50	0.0054	
	> 0.50 to 0.80	0.0054	
	> 0.80 to 0.99	0.0053	
601 nm to 1500 nm	up to 0.02	0.0017	
	> 0.02 to 0.05	0.0022	
	> 0.05 to 0.10	0.0025	
	> 0.10 to 0.20	0.0052	
	> 0.20 to 0.50	0.0064	
	> 0.50 to 0.80	0.0064	
	> 0.80 to 0.99	0.0049	
1501 nm to 2200 nm	up to 0.02	0.0090	
	> 0.02 to 0.05	0.0090	
	> 0.05 to 0.10	0.015	
	> 0.10 to 0.20	0.015	
	> 0.20 to 0.50	0.0099	
	> 0.50 to 0.80	0.0083	
	> 0.80 to 0.99	0.0088	
2201 nm to 2500 nm	up to 0.02	0.054	
	> 0.02 to 0.05	0.054	
	> 0.05 to 0.10	0.043	
	> 0.10 to 0.20	0.043	
	> 0.20 to 0.50	0.035	
	> 0.50 to 0.80	0.028	
	> 0.80 to 0.99	0.032	
END			

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Notes

Note 1: A Calibration and Measurement Capability (CMC) is a description of the best result of a calibration or measurement (result with the smallest uncertainty of measurement) that is available to the laboratory's customers under normal conditions, when performing more or less routine calibrations of nearly ideal measurement standards or instruments. The CMC is described in the laboratory's scope of accreditation by: the measurement parameter/device being calibrated, the measurement range, the uncertainty associated with that range (see note 3), and remarks on additional parameters, if applicable.

Note 2: Calibration and Measurement Capabilities are traceable to the national measurement standards of the U.S. or to the national measurement standards of other countries and are thus traceable to the internationally accepted representation of the appropriate SI (Système International) unit.

Note 3: The uncertainty associated with a measurement in a CMC is an expanded uncertainty using a coverage factor, $k = 2$, with a level of confidence of approximately 95 %. Units for the measurand and its uncertainty are to match. Exceptions to this occur when marketplace practice employs mixed units, such as when the artifact to be measured is labeled in non-SI units and the uncertainty is given in SI units (Example: 5 lb weight with uncertainty given in mg).

> **Note 3a:** The uncertainty of a specific calibration by the laboratory may be greater than the uncertainty in the CMC due to the condition and behavior of the customer's device and specific circumstances of the calibration. The uncertainties quoted do not include possible effects on the calibrated device of transportation, long term stability, or intended use.
>
> **Note 3b:** As the CMC represents the best measurement results achievable under normal conditions, the accredited calibration laboratory shall not report smaller uncertainty of measurement than that given in a CMC for calibrations or measurements covered by that CMC.
>
> **Note 3c:** As described in Note 1, CMCs cover calibrations and measurements that are available to the laboratory's customers under *normal conditions.* However, the laboratory may have the capability to offer special tests, employing special conditions, which yield calibration or measurement results with lower uncertainties. Such special tests are not covered by the CMCs and are outside the laboratory's scope of accreditation. In this case, NVLAP requirements for the labeling, on calibration reports, of results outside the laboratory's scope of accreditation apply. These requirements are set out in Annex A.1.h. of NIST Handbook 150, Procedures and General Requirements.

Note 4: Uncertainties associated with field service calibration may be greater as they incorporate on-site environmental contributions, transportation effects, or other factors that affect the measurements. (This note applies only if marked in the body of the scope.)

Note 5: Values listed with percent (%) are percent of reading or generated value unless otherwise noted.

Note 6: NVLAP accreditation is the formal recognition of specific calibration capabilities. Neither NVLAP nor NIST guarantee the accuracy of individual calibrations made by accredited laboratories.

Note 7: See NIST Handbook 150 for further explanation of these notes.